1.0 PURPOSE

To outline management policy for

- Operational planning and control
- Requirements for products and services
- Design and development of products and services
- Control of externally provided processes, products and services
- Production and service provision
- Release of products and services
- Control of nonconforming output

2.0 SCOPE

Applicable to all products & processes at NIHVA

3.0 DETAILS

3.1 Operational planning and control

3.1.1 The planning of controls is required to ensure consistent acceptability of products and services. Planning processes include the definition of quality objectives, development for required processes, establishment for appropriate verification programs and the requirement for records necessary to demonstrate the process and products conform to intended requirements. NIHVA plans, implements and controls the processes needed to meet the requirements for the provision of products and services, and to implement the actions determined in Planning by:

- Determining the requirements for the products and services
- Establishing criteria for the processes and the acceptance of products and services
- Determining the resources needed to achieve conformity to the product and service requirements
- Implementing control of the processes
- Determining, maintaining and retaining documented information to the extent necessary to have confidence that the processes have been carried out as planned, and to demonstrate the conformity of products and services to their requirements

The output of operational planning and control includes documented customer inputs, resource requirements, processes, equipment requirements, procedures, installation/ Project data, and design outputs.

Operational Planning and Control



3.1.2 Quality planning includes following activities: -

- Technology, equipment, processes & skills required to achieve desired quality are identified and acquired.
- Compatibility of design, production processes, inspection & test procedures and related documentation is ensured.
- Quality records are identified, prepared and maintained.
- The updating, as necessary of quality control, inspection and testing techniques, including the development of new instrumentation.
- The identification of any measurement requirement.
- The identification of suitable verification at appropriate stages in the realization of product
- The clarification of standards of acceptability for all features and requirements, including those that contain a subjective element.

3.1.3 Control over outsourced processes: - In case NIHVA chooses to outsource any process that affects product conformity with requirements, it is ensured that appropriate control is exercised over such processes. The type and extent of control to be applied to outsourced process can be influenced by factors such as

- The potential impact of outsourced process on NIHVA's capability to provide product that conforms to requirements
- The degree to which the control for the process is shared
- The capability of achieving the necessary control

The outsourced processes at NIHVA for its QMS are –

- Calibration (by outside agencies)
- Subcontracted activities for commissioning

The controls exercised on these processes are:

- Calibration: - Ensure traceability.
- Subcontracted activities – adequate controls are exercised through supervision by NIHVA

3.2 Requirements for products and services

3.2.1 Customer Communication includes,

- Providing information related with products and services
- Handling enquiries, contracts or orders including changes
- Obtaining customer feedback related with products and services, including customer complaints
- Handling or controlling customer property
- Establishing specific requirements for contingency actions, when relevant.

3.2.2 Customer communication: - Procedures and Responsibilities are established within marketing department for effective communication with customer in relation to product information, inquiry, contract or order handling, order amendments, customer feedback including complaints. The primary means of communication are defined in the referenced procedures; however, at any time other ad hoc methods may be used as appropriate for a given business situation.

3.2.3 Determining the requirements for products and services: NIHVA has established processes to ensure that the requirements for products and services are defined.

Following requirements are determined:

- Requirements specified by the customer, including delivery and post-delivery activities. The post-delivery activity include actions under warranty provisions, contractual obligations such as maintenance services, and supplementary services such as recycling or final disposal.
- Requirements not stated by the customer but necessary for specified use or known intended use;
- claims for products and services
- Statutory and regulatory requirements applicable to the product; and
- Any additional requirements considered as necessary by NIHVA.

3.2.4 Review of Requirements Related to Products and Services

NIHVA Technologies ensures we have the ability to meet the requirements for products and services to be offered to customers. Management conducts a contract/product review prior to committing to supply products and services to a customer. The review process at a minimum includes:

- requirements specified by the customer, including the requirements for delivery and post- delivery activities;
- requirements not stated by the customer, but necessary for the intended use, when known;
- requirements specified by the organization;
- statutory and regulatory requirements applicable to the products and services
- Contract or order requirements differing from those previously expressed.

NIHVA Technologies ensures contracts, purchase orders or other requirements differing from those previously defined, are reviewed and approved prior to incorporating into our business systems.

We retain applicable documented information of the initial review and on any new/revised customer or applicable external party requirements for the products and services provided.

3.2.5 Changes to Requirements for Products and Services

NIHVA Technologies ensures that relevant documented information is amended, and that relevant persons are made aware of the changed requirements, when the requirements for products and services are changed.

3.3 Design and development of products and services

NIHVA Technologies performs basic design activities for material handling equipment (Scissor lift, dock leveler) and door automation (Gates, sliding roof and rolling shutters). NIHVA does not perform design activities for other traded items.
NIHVA has established, implemented and maintained a process for design and development that is appropriate to ensure the subsequent provision of products and services

3.3.1 Design and development planning:

In determining the stages and controls for design and development, NIHVA Considers-

- The nature, duration and complexity of design and development activities
- The required process stages, including applicable design and development reviews
- The required design and development verification and validation activities
- The responsibilities and authorities involved in the design and development process

- The internal and external resource needs for the design and development of products and services
- The need to control interfaces between persons involved in the design and development process
- The need for involvement of customers and users in the design and development process
- The requirements for subsequent provision of product and services
- The level of control expected for the design and development process by customers and other relevant interested parties
- The documented information needed to demonstrate that design and development requirements have been met

3.3.2 Design and development inputs

3.3.2.1 Relevant inputs to the design and development activities are determined and reviewed for adequacy to ensure that these inputs are clear and unambiguous. These inputs include -

- Functional performance requirements
- Statutory and regulatory requirements if any
- Information from previous similar designs where applicable
- Any other requirements essential for design and development.

3.3.2.2 Design input / specifications get generated as a result of contract review activity. Procedure is established to determine the requirements essential for the specific types of products and services to be designed and developed. Consideration is given to -

- Functional and performance requirements
- Information derived from previous similar design and development activities
- Statutory and regulatory requirements
- Standards or codes of practice of Eaton
- Potential consequences of failure due to the nature of the products and services

3.3.2.3 Procedure is established for identification and review of process related design inputs such as

- Customer requirements if any, regarding process
- Learning from previous similar projects
- Design output data

3.3.3 Design and development controls: Processes are established to apply controls to the design and development which ensures-

- The results to be achieved are defined
- Reviews are conducted to evaluate the ability of the results of design and development to meet requirements
- Verification activities are carried out to ensure that the resulting products and services meet the requirements for the specified application or intended use
- Any necessary actions are taken on problems determined during the reviews, or verification and validation activities
- Documented information of these activities in retained

3.3.3.1 Design and development review - reviews are performed during design and development process. The goals of these reviews are to:

- Evaluate the ability of the results of design and development to meet identified requirements

- Identify any problems and determine necessary actions

3.3.3.2 Design and development verification: - design verification is carried out to ensure that the design output meets design input requirements. Design verification is in the form of checking of Design input and design output (design calculations & drawings)

3.3.3.3 Design and development validation: -

- Concerned sales / project team submits the report pertaining to satisfactory working of the products. Such reports will be considered as validation records
- In case of repeat designs, re-validation is not required
- Product design provided by customer - if no complaint is received from the customer during warranty, the design is considered to be valid
- Products designed as per specifications from customer: - if no complaint is received from the customer during warranty, the design is considered to be valid.

3.3.4 Design and development outputs

The outputs of design and development include -

- Verify that design output meets design input requirements
- Are adequate for the subsequent processes for the provision of products and services.
- Include or reference monitoring and measuring requirements, as appropriate, and acceptance criteria
- Specify characteristics of the products and services that are essential for intended purpose and their safe and proper provision
- Design output is documented in form of drawing and bill or material

3.3.5 Design and development changes: - Design and development changes are identified and records maintained. The changes are reviewed, verified and validated, as appropriate, and approved before implementation through Engineering Change Note (ECN). Records of the review of changes and any necessary actions is maintained

3.4 Control of externally provided processes, products and services:

NIHVA Technologies maintains responsibility for the quality of all products purchased from external providers, including customer designated sources. Procedures ensure products and services being provided by external sources will conform to our customers' requirements. Examples of our controls include:

- A documented Approved Supplier List
- The review of external provider's performance

3.4.1 Type and extent of control: NIHVA has established a Procedure to ensure that externally provided processes, products and services do not adversely affect the ability to consistently deliver conforming products and services to customers.

Purchasing procedures are established to

- Ensure that externally provided processes remain within the control of its QMS
- Define both the controls that it intends to apply to an external provider and those it intends to apply to the resulting output

- Take into consideration, the potential impact of the externally provided processes, products and services on the organization's ability to consistently meet customer and applicable statutory and regulatory requirements
- Take into consideration, the effectiveness of the controls applied by the external provider
- Determine the verification, or other activities necessary to ensure that the externally provided processes, products and services meet requirements.

Verification of purchased product: -

- Procedures are defined for receiving inspection activities.
- Where NIHVA or its customer intends to perform verification at the supplier's premises, the intended verification arrangements and method of product release are specified in the purchasing information.

3.4.2 Information for external providers: Purchasing processes are established to ensure the adequacy of requirements prior to their communication to the external provider.

Purchase order describes the product or service to be purchased including where applicable, following requirements.

- The processes, products and services to be provided
- Requirements for approval of products and services / methods, processes and equipment / release of product and services
- Requirements for competence, including any required qualification of persons
- The external providers' interactions
- Control and monitoring of external providers' performance to be applied
- Requirements about verification or validation activities
- Quality management system requirements.

3.5 Control of Production and service provision

NIHVA Technologies plans and implements production and service provision and as required by job specific requirements. Manufacturing/ Assembly procedures, AMC Contracts, inspection plans, and other documents deemed necessary, define the acceptance for manufacturing and service operations. The plans provide detailed instruction and guidance for all production and service phases including the methods and equipment to be used. Records for each sales order of product produced/ install/ commission provide unique traceability. This record is maintained as required by customer contract requirements

3.5.1 A procedure is established to plan and carry out production and service provision under controlled conditions. Controlled conditions includes the following as applicable

- The availability of documented information that defines the characteristics of the product to be produced, the services to be provided, or the activities to be performed and the results to be achieved
- The availability and use of suitable monitoring and measuring resources
- The implementation of monitoring and measurement activities at appropriate stages to verify that criteria for control of processes or outputs, and acceptance criteria for products and services, have been met
- The use of suitable infrastructure and environment for the operation of processes
- The appointment of competent persons including any required qualification
- The validation and periodic revalidation of the ability to achieve planned results of the processes for production and service provision, where the resulting output cannot be verified by subsequent monitoring or measurement

- The implementation of actions to prevent human error
- The implementation of release, delivery and post-delivery activities.

3.5.1.1 Procedure is established to validate any processes for production and service provision where the resulting output cannot be verified by subsequent monitoring or measurement. This includes any processes where deficiencies become apparent only after the product is in use or the service has been delivered. Following processes have been identified as special processes

- Welding
- Surface treatment processes like Plating

3.5.1.2 Arrangements are established for these processes including, as applicable

- Defined criteria for review and approval of the processes,
- Approval of equipment and qualification of personnel,
- Use of specific methods and procedures,
- Requirements for records
- Revalidation.

3.5.1.3 Operators are qualified by test or experience

3.5.2 Identification and traceability: -

- NIHVA Technologies identifies parts and products by suitable means for supplies to customer. According to the level of traceability required by contract, regulatory or other established requirement, our procedures provides for:
- identification to be maintained throughout the processes including delivery and post-delivery;
- Identification of sub-components and those of the next higher assembly.
- Procedures and responsibilities are defined for identification of the inspection status through all phases of production.
- Inspection and test status is identified either by tags or by storage locations that are clearly identified or by a document or on the material itself.
- When required by the customer, additional verification and identification requirements are met.

3.5.3 Property belonging to Customers or external providers

- In case of customer / external provider supplied property (parts or intellectual property); procedure and responsibilities are established for Receiving, Receiving inspection, Marking, Handling, Storage and Maintenance.
- Procedure is established to identify, verify, protect and safeguard customers' or external providers' property provided for use or incorporation into the products and services
- Any such property which is lost, damaged or otherwise unsuitable for use is recorded and reported to the customer / external provider
- Production tooling owned by customer / external provider (if available) are marked permanently

3.5.4 Preservation

- NIHVA Technologies preserves the conformity of parts and products during internal processing and delivery to the intended destination including outside services. Procedures include instructions for identification, handling, packaging, storage and protection. Preservation of outputs also includes, where applicable:

- cleaning;
- prevention, detection and removal of foreign objects;
- special handling for sensitive outputs;
- marking and labeling including safety warnings;
- Special handling for hazardous materials.

- The shipping department ensures that documents required by the contract/order to accompany the product are present at delivery and are protected against loss and deterioration

3.5.5 Post-delivery activities: -

- While establishing procedure for post-delivery activities, NIHVA considers statutory and regulatory requirements, potential undesired consequences associated with its products and services, nature / use and intended lifetime of products and services, customer requirements and customer feedback.
- Responsibilities and procedures for product service are defined. Servicing activity is coordinated by service department

3.5.6 Control of Changes: -

NIHVA Technologies shall review and control changes to the extent necessary to ensure continuing conformity of customer or internal requirements. Changes may be initiated as a result of:

- needs of interested parties, or customer feedback;
- Opportunities for improvement are identified.

Management reviews and monitors changes that affect project or outside services and ensures change documentation and information is distributed and controlled. Records of results of the review of changes, the persons authorizing the change, and any necessary actions arising from the review are maintained in accordance with applicable procedures.

3.6 Release of products and services:

NIHVA Technologies monitors and measures the characteristics of the product in receiving inspection, in- process inspection, and final inspection to verify that requirements have been met. Documented procedures have been established for product inspection. Documented Records and information of inspection include evidence of conformity with the acceptance criteria and traceability to the person authorizing the release. Records of inspection are maintained

3.7 Control of nonconforming outputs

- Procedures and responsibilities are established to ensure that outputs which does not conform to their requirements are identified and controlled to prevent its unintended use or delivery.
- A procedure is established to define the controls and related responsibilities and authorities for dealing with nonconforming output. Actions are initiated based on nature of the conformity and its effect on the conformity of products and services. This also applies to nonconforming products and services detected after delivery of products, during or after the provision of services
- Nonconforming products are identified and stored separately
- Nonconforming outputs are dealt with one or more of following ways
  - Correction
  - Segregation, containment, return or suspension of provision of products and services
  - Informing the customer
  - Obtaining authorization for acceptance under concession

- Conformity to the requirements are verified when nonconforming outputs are corrected
- Records of nonconformities are maintained as required and include:
  - description of nonconformity;
  - description of actions taken;
  - description of concessions obtained;
  - Identification of the authority deciding the action in respect of the nonconformity.
- When nonconforming product is corrected, it is re-inspected to the original specifications and requirements to ensure it conform to customer stated requirements. When a nonconforming product is detected after delivery, NIHVA Technologies will take action appropriate to the effects or potential effects of the nonconformity.

3.8 Related Documents

QSP-07-001 Control of Documents & Information	QSP-08-006 Vendor Evaluations
QSP-08-001 Control of Nonconformance's	QSP-08-007 Inspection
QSP-08-002 Engineering Change Notice	QSP-08-010 Preservation
QSP-08-003 Identification & Traceability	QSP-08-012 Contract Review
QSP-08-004 Rework	QSP-08-014 Shipping& Receiving
QSP-08-005Purchasing	QSP-08-015 Control of Customer Property